

RESEARCH PRESENTATION · URBAN STREETSCAPE × COMMERCIAL DISTRICTS

What Street View Can — and Cannot — Sense About Commercial Districts

A Quantitative Capability Map of a Selective Visual Sensor

825 Seoul commercial districts · Google Street View (POI-anchored) + DINOv2 · Cross-sectional

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1 Where We Started — Does Streetscape Relate to Sales?

A long-standing question in urban planning and retail-district analysis

Motivation

- Does a “better-looking” street go with a stronger commercial district? A classic question in urban studies.
- We wanted to feed street images into a model, compare them against district sales, and learn which visual aspects of a street relate to its sales.
- Street View + deep learning let us turn “how a street looks” into numbers we can test.

Initial design — an A/B comparison

A · Baseline

Predict sales from structural / tabular variables only (workplace pop., residents, stores, area, type).

B · Multimodal

Baseline + street-image features. Does the image add explanatory power on top of structure?

This A vs B logic survives all the way to RQ3.

2 First Pivot — The Time-Series Baseline Was Too Weak

We changed the baseline model because it underperformed, not merely because of COVID

What we built — a Panel-LSTM (RNN) baseline

- Panel LSTM: 4-quarter lookback, 64 hidden units, 2 layers ($\approx 56K$ params).
- Inputs: time-varying X (floating/workplace/resident pop., stores, open/close rates, CPI, rate, seasonality) + static (type, area) + district embedding.
- Lag of past sales deliberately excluded — to isolate whether images add power over structure.
- Train 22Q2–24Q1 · Val 24Q2–Q4 · Test 25.

It lost to a naive baseline

LSTM $R^2 = 0.64$ vs.

Naive (per-district mean) $R^2 = 0.94$

A simple historical average beat the LSTM (sMAPE 87% vs 18%).

Why — and what it told us

Most explainable variance lives between districts, not over time. The temporal signal was weak (and COVID made the series noisier). So we moved the baseline to a cross-sectional design.

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Second Pivot — From Causation to Capability

Why a single-snapshot, cross-sectional design cannot identify causal effects



Causation — not identifiable here

- District type, location and demographics co-vary with streetscape — in one snapshot they cannot be separated from the streetscape's effect.
- Street View is a single snapshot, so the same district can't be tracked over time.
- Evidence: once we control for structure in RQ3, the image's added effect shrinks sharply.



Capability — what we can ask

Reframed question

What can Street View **detect** about a district, and what can it **not**? Where does that ability come from, and does anything remain for sales after controls?

Tone of the talk Honesty over over-claiming — we map both what the tool can do and where it stops.

4 Three Questions — The Roadmap

RQ1

Capability map

From street images alone, which structural variables can we recover?

→ What it sees

RQ2

Mechanism

Where does that ability come from — and can we interpret the black box?

→ Why & how it sees

RQ3

Boundary

After controlling for demographics & type, do images still explain sales?

→ Whether it's useful

Preview The value comes from RQ1 & RQ2; RQ3 marks the boundary of that ability.

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Data & Image Collection (POI-Anchored)

All public data · 14,284 images · cameras aimed at storefronts

Data sources

- **Seoul Commercial Analysis**

sales · floating/workplace/resident pop. · stores · open-close

- **Small Enterprise Agency**

individual store POI coordinates

- **Google Street View**

street panoramas (pre-2019 vintage)

- **District boundaries**

commercial-area polygons / wards

What is a POI? a Point of Interest = one individual store's coordinate.

- We aim the camera at the storefront (heading auto-computed from the store point).
- The older $\pm 90^\circ$ (road-perpendicular) method captured mostly road/sky in alleys — POI anchoring fixes this structurally.

Signal gain jobs R^2 0.47→0.55, type AUC 0.86→0.94

Sampling scaled to district size

- Stores $\geq 20\text{m}$ apart are spatially spread across the polygon — larger districts get wider coverage.
- Up to 30 points queried per district; only pre-2020 panoramas kept (blocks reverse causality) → ~16 images each.
- 16,362 collected (≤ 2019) → off-vintage pooled & back-filled → 31 sparse districts (<10 imgs) dropped → 14,284 used / 825 districts (~17 each).

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Defining the Study Sample

Two district types · 825 districts total

Developed districts n=202

Large districts on main roads / transit hubs (e.g., Teheran-ro, Yeouido)



Alley districts n=623

Small neighborhood (back-street) districts near residential areas



Exclusions — shared reason: street-level commercial scene is absent or inappropriate

Traditional markets / tourist zones

Area < 50,000 m²

Indoor complexes (malls, underground)

Transit hubs (airport, SRT)

Industrial / wholesale

Parks · campuses

Apartment complexes · hospitals

Sales < ₩100M

Districts with < 10 images

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RQ1 Design — Predicting Structure From Pixels Only

“Image-only” = no tabular/administrative data; the street photo is the sole input



X = image features only → **Y = one structural variable (workplace/resident/floating pop., stores, F&B, open/close ... / type = classification)**

What we read from LASSO

- We ask: from image features alone, how much of a variable can a sparse linear model recover?
- Fit on training folds, predict held-out districts, score R^2 — repeated in cross-validation and averaged.
- R^2 here = the share of a variable’s variance the image explains on unseen districts = its detectability.

The three regressors — and why LASSO

LASSO — L1 penalty, drives unhelpful weights to exactly 0.

Ridge — L2, shrinks all weights smoothly.

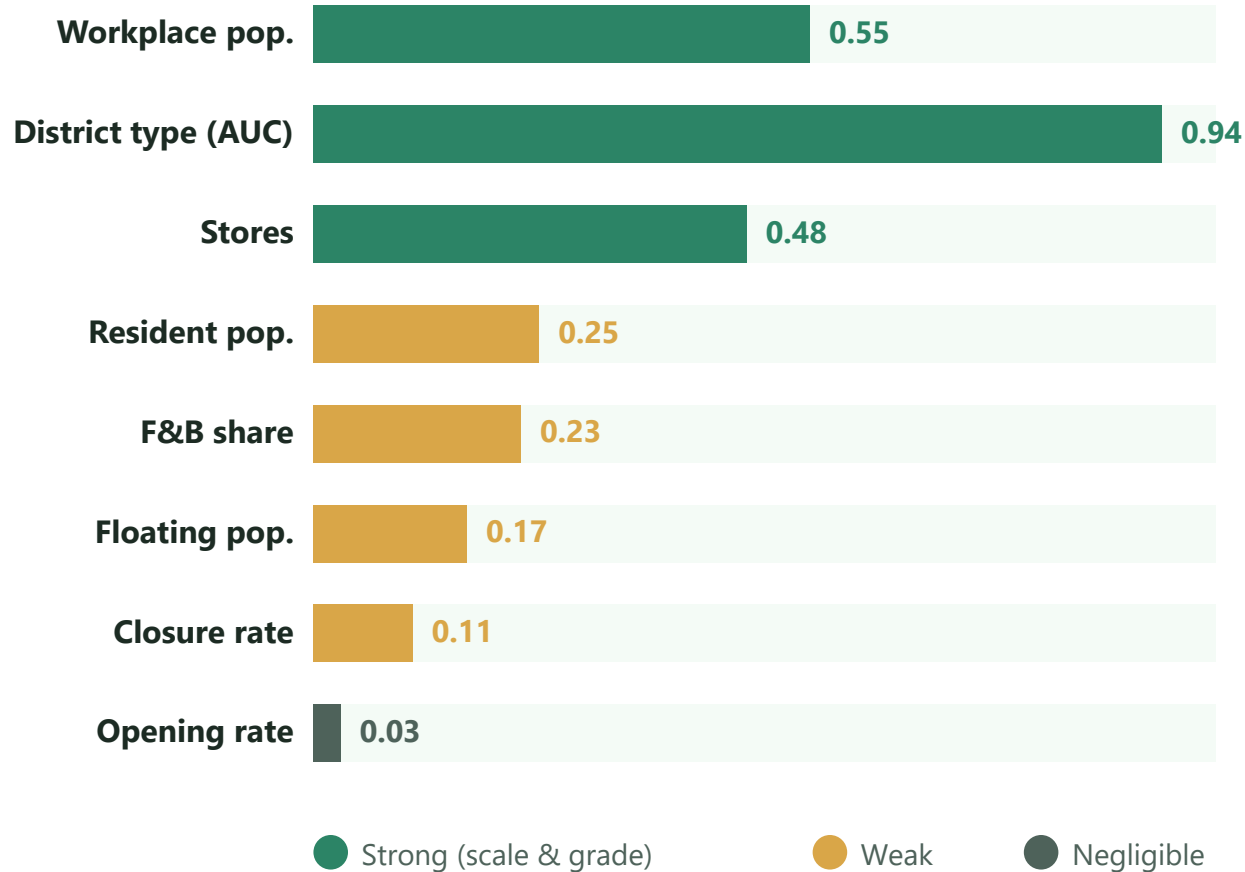
ElasticNet — a blend of L1 and L2.

Robustness: all three give near-identical R^2 (0.546 / 0.556 / 0.546) — not a quirk of one model.

Why LASSO as the main model: it zeros out unneeded features, keeping only the few that matter — sparse, interpretable, and safe against overfitting.

RQ1 Results — The Capability Map

Image-only (no admin data) · cross-validated R^2 / AUC



Reading it

Strong: commercial scale & grade — workplace pop., stores, district type.

Weak: resident pop., floating pop., business mix, churn.

Why AUC for type? type is a binary category (developed vs alley), not a continuous number — so we score classification (AUC), still on a 0–1 scale.

A Selective Sensor — Not Just Memorizing “Busy Places”

Workplace population (structural)

0.55

detected strongly ✓

Floating pop. (the most direct “busyness”)

0.17

barely detected ✕

The argument If the model had simply memorized “big, crowded places,” it would nail floating population — the most direct proxy for busyness (people passing by). It doesn’t (0.17). Instead it recovers commercial structure (workplace pop., stores). It reads the built form, not raw crowd size.

Bridge Images also predict sales ($\log\text{-sales } 0.51 \cdot \text{sales} \div \text{floating pop. } 0.54$) — but does that survive controlling for structure? → RQ3.

Selectivity Holds Even Within a Single Type

Group	Workplace pop.	Stores	Resident pop.	Floating pop.	Type (AUC)
Pooled (n=825)	0.546	0.478	0.246	0.173	0.937
Alley (n=623)	0.176	0.244	0.270	0.223	—
Developed (n=202)	0.315	0.012	-0.073	-0.010	—

What this means

Much of the pooled selectivity comes from the developed-vs-alley difference. But inside developed districts, **only workplace population (0.315) survives** — store count, resident and floating population collapse to ~0 or below.

The model senses even fine differences in office density, beyond the binary type.

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Opening the Black Box — Why Only Two Targets

DINOv2's 768 numbers don't tell us what the model looked at; we decompose only the two it senses strongly

Why decompose only Workplace pop. & Stores?

These are the two variables the image senses strongly ($R^2 \geq 0.48$). Only there is enough signal to take apart — decomposing a weak target (e.g., floating pop. 0.17) would just split noise.

DINOv2 = 768 numbers

???

we can't see what it looked at

1

Human-readable parts — Segmentation

Building, sky, road, greenery ... read as area shares.

2

Quantify the signal — Commonality

Split prediction into parts that reduce to readable features vs. parts unique to the deep model.

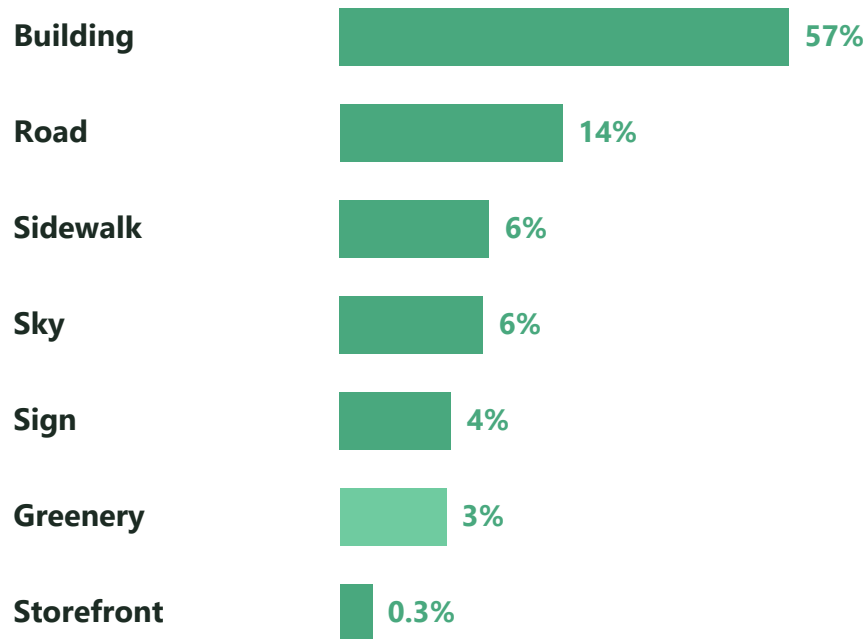
10 RQ2① Segmentation — Method & Overall Composition

How a “57% building” number is produced

How we use segmentation

- Each photo is classified pixel-by-pixel into scene classes (ADE20K labels; SegFormer model, pretrained).
- For one photo → share of each class (e.g., building pixels ÷ all pixels).
- Average across a district’s ~16 photos → district composition.
- Average across the 825 districts → the overall figures at right (e.g., building ≈ 57%).
- Readable, unlike the DINO black box: it shows what is in the scene.

Overall street composition (mean of 825 districts)



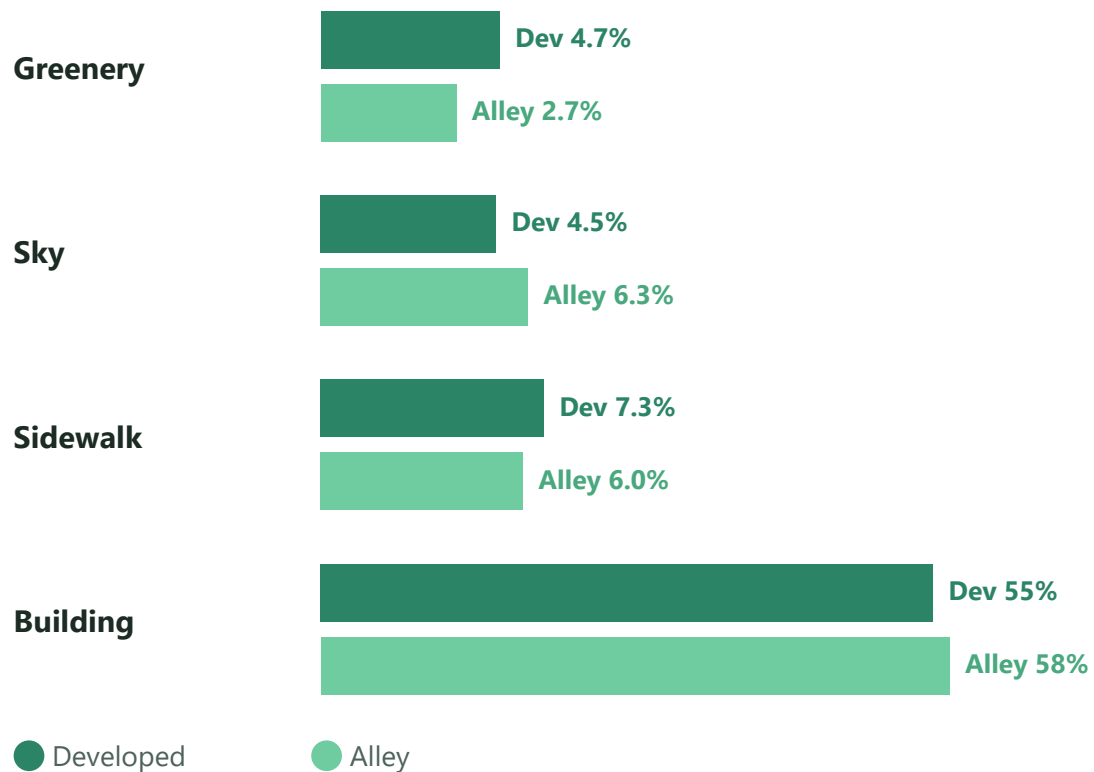
Storefronts are just 0.3% — yet RQ1 reads commercial structure well, so the model relies on building form, signage and road layout, not storefronts themselves.

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RQ2① Developed vs. Alley — What Differs

Composition by type, and how each element correlates with workplace pop.

Mean composition by district type



Correlation with workplace pop. (by PCC)

Greenery	+0.33
Sidewalk	+0.22
Sign	-0.17
Sky	-0.26
Building	-0.12

Workplace population is encoded as "enclosure (low sky = tall buildings) + wide sidewalks + street trees"; signage marks retail/alley character.

Greenery Was Not “Quietness”

Greenery share — Developed vs Alley

0.047 Developed

0.027 Alley $\approx 2\times$ more

Correlation with workplace pop. $r = +0.33$



What it really is The greenery the model sees is not rural nature but the street trees and landscaping of dense office boulevards. **The model picked up a Seoul-specific pattern that contradicts the human “green = quiet” intuition.**

12 RQ2② Handcrafted Features — Seeing What Shares Miss

Why area ratios alone are not enough

Why add them?

- Segmentation gives only how much of each class is present (area share).
- It cannot describe texture, pattern, regularity, or color — two streets with the same “60% building” can look completely different.
- Part of the DINO signal that segmentation can’t explain might just be low-level visual statistics, not something mysterious.

What handcrafted features capture

- **GLCM** — texture roughness / regularity
- **Gabor** — oriented / directional texture
- **Edge density** — visual complexity
- **HSV color** — color statistics
- **Entropy** — scene disorder

Why it matters These ~16 classic computer-vision features are defined by formulas. If they absorb part of DINO’s “unexplained” signal, we’ve shown that signal is interpretable low-level vision — not a mystery.

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RQ2③ Splitting the Signal — Commonality Analysis

Three feature sets added in steps; each step's R^2 gain = its unique contribution

Cumulative R^2 as we add each set (target = Workplace pop.)

Segmentation only

composition / area shares



+ Handcrafted texture

GLCM · Gabor · edge · color



+ DINO (black box)

768-dim deep features



Texture absorbs +0.15 ($\approx 28\%$); DINO still adds +0.15 ($\approx 28\%$) that no readable feature explains.

How each step is fit

Same regressor, same Y (Workplace pop., then Stores); only X changes:

Seg = area shares; **Texture** = 16 handcrafted stats; **DINO** = 768 deep features.

What the gains mean

Seg overlaps with DINO as the interpretable 'shared' part. If handcrafted texture absorbs part of DINO's gain, that part is just low-level vision; whatever still remains is genuinely DINO-unique.

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RQ2③ Results — 72% of the Signal Is Interpretable

Both strongly-sensed targets decompose almost identically

Workplace pop. ($R^2=0.54$)



Stores ($R^2=0.48$)



What the split means

72% interpretable: reducible to scene composition + classic texture/color — things people can name.

28% DINO-unique: not reducible to any area share or texture statistic — the spatial arrangement / architectural gestalt only the large vision model captures.

Two doubts people raise about black-box ViTs — and our answer

“Did it just learn random noise?” → structured texture absorbs ~28%, so the signal is not random.

“Did it only look at building area?” → composition is only 44%; most of the signal is elsewhere.

15 RQ3 Design — Does the Image Add Anything for Sales?

Baseline vs. multimodal, and why these choices

The comparison

Baseline (A): predict sales from structure only — log workplace/resident/floating pop. + area + ward fixed effects + district type (LASSO).

Multimodal (B): baseline + image features (DINO→PCA, top components correlated with sales).

$\Delta R^2 = B - A$ = what the image adds beyond structure.

Why $Y = \text{sales} \div \text{floating pop.}$

Total sales is dominated by scale (a big, busy district sells more). Dividing by floating population (people passing by) gives sales per passer-by = conversion efficiency, removing the volume that location/scale already drive. We also report log-sales as a check.

Why these controls

Workplace/resident/floating pop., area, ward and type are exactly what the image already proxies (RQ1). Putting them in the baseline asks the strict question: does the image add information beyond the structure it encodes? We test pooled and within each type.

16 RQ3 Results — A Boundary, Mostly a Proxy

Increment is small and significant only in alley districts

Level	Y (outcome)	Baseline	+ Image (MM)	ΔR^2	p
Pooled	log sales (total)	0.686	0.729	+0.043	0.003 **
Pooled	sales ÷ floating pop.	0.609	0.659	+0.049	0.004 **
Alley	log sales	0.382	0.464	+0.082	<0.001 ***
Alley	sales ÷ floating pop.	0.245	0.349	+0.104	0.001 ***
Developed	log sales	0.502	0.554	+0.052	0.190 n.s.
Developed	sales ÷ floating pop.	0.211	0.254	+0.043	0.521 n.s.

How to read it The increment is small; in developed districts it is not significant (p=0.190, 0.521) — the image is largely a proxy for structure. Only alley districts show a robust signal worth a future causal test. $\Delta R^2 > 0$ is association, not causation.

17 Synthesis — What This Means

Street View is best understood as a strong visual proxy for a district's commercial structure — a selective sensor, read honestly with its limits.

Significance for practice

Sensing workplace pop. & store density from imagery alone suggests visual estimation of commercial structure where administrative data is scarce or out of date.

Significance for method

A reusable way to open a vision black box — separating prediction into readable composition/texture vs. model-unique signal.

Significance for data design

How images are sampled (POI-anchored) materially changes what can be sensed.

In short the contribution is the map of ability and limits — not the size of the RQ3 increment.

Limitations & Future Work — Toward Causation

Cross-sectional = association; here is the path to a causal answer

Limitations

- A single-snapshot, cross-sectional design cannot identify causal effects.
- One image vintage; features aggregated to the district level.
- Sample is consumer districts only (industrial areas excluded) — read the map within that scope.
- The alley-district signal is a lead to verify, not proof.

A causal design — Difference-in-Differences

1

Treatment

Use street-improvement projects (e.g., pedestrian-environment districts) as an external intervention.

2

Before / after

Compare sales change in treated vs. untreated districts, before vs. after, with road-view of both periods.

3

Role of images

Not the identification tool itself — images measure and verify the visual “dose” of the treatment.

CONCLUSION

We set out to learn what a street image can tell us about a district —

and we mapped, both what it can and cannot sense.

RQ1 workplace pop. $R^2=0.55$ · type AUC=0.94 | RQ2 interpretable 72% + unique 28% | RQ3 increment small, alley-only

Thank you.